

GLASGOW COLLEGE UESTC

Exam paper

Electromagnetic Field and Microwave Technology (4002)

Date: June 19th 2013

Time: 14:30-16:30PM

Attempt all PARTS. Total 100 marks

Use one answer sheet for each of the questions in this exam.

Show all work on the answer sheet.

Make sure that your University of Glasgow and UESTC Student Identification Numbers are on all answer sheets.

An electronic calculator may be used provided that it does not allow text storage or display, or graphical display.

All graphs should be clearly labelled and sufficiently large so that all elements are easy to read.

The numbers in square brackets in the right-hand margin indicate the marks allotted to the part of the question against which the mark is shown. These marks are for guidance only.

DATA/FORMULAE SHEET IS PROVIDED AT THE END OF PAPER

- Q1 (a) What is the static electric field? Explain why the static electric field in a conducting body is zero. [4]
- (b) State the differential equation of static electric scalar potential in source free region. What are the boundary conditions of static electric scalar potential on the interface of two lossless medium with permittivity of ϵ_1 and ϵ_2 ? [6]
- (c) Explain the definition of capacitance. Calculate the capacitance of an isolated conducting sphere with radius of a in air. [6]
- (d) The interface for two media is xoz plane, the magnetic fields on the interface are $\vec{H}_1 = 3\vec{e}_x + 4\vec{e}_y + 7\vec{e}_z (y = 0_+)$ and $\vec{H}_2 = -3\vec{e}_x + 2\vec{e}_y - 3\vec{e}_z (y = 0_-)$, what is the surface current density $\vec{J}_s$ on the interface. Explain the difference between conduction current and magnetized current. [6]
- Q2 (a) State the definition of time harmonic electromagnetic field and the motivation to study time harmonic electromagnetic field. [4]
- (b) Illustrate the Poynting theorem in differential form and its physical meaning [4]
- (c) Explain what is the polarization of a uniform plane waves, and the different types of polarization? Illustrate the type of the polarization for the following uniform plane wave, $\vec{E} = \vec{e}_x 10e^{j2\pi z} - j\vec{e}_y 10e^{j2\pi z}$. [8]
- (d) A long straight wire and a loop are placed inside the open space as shown in **Figure Q2**, the wire is at the interface of air (μ_0) and magnetic material (μ) where loop is located. Determine the magnetic flux density $\vec{B}$ and the mutual inductance between the wire and the loop. [12]

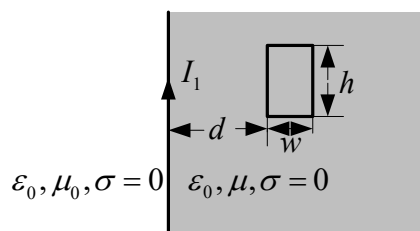


Figure Q2

Continued overleaf

Q3

A uniform plane wave is propagating in a medium with $\epsilon_r = 8, \mu_r = 2$. Its electric field intensity is expressed as $\vec{E} = \vec{e}_x E_0 e^{j2z}$. Please solve the following:

- The propagation direction of the plane wave; [2]
- The wavelength, frequency and angular frequency of the plane wave; [4]
- The magnetic field intensity of the plane wave; [4]
- Consider the perfect conducting boundary shown in **Figure Q3**. When the uniform plane wave is normally incident on the boundary of a perfect conductor, find the total electric field and average poynting vector in the region $z > 0$. [10]

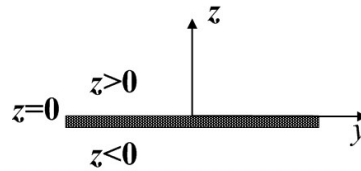
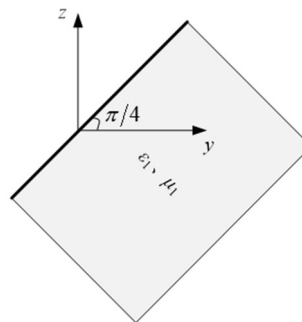


Figure Q3

Q4

A uniform plane wave is propagating in free space. Its electric field intensity is expressed as $\vec{E} = \vec{e}_x E_0 e^{-j(y-z)}$. Please solve the followings:

- The unit vector along the propagation direction of the plane wave; [5]
- The wave number and period of the plane wave; [6]
- The average Poynting vector of the plane wave; [7]
- Consider the perfect dielectric boundary shown in Figure Q4. The angle between the boundary and y-axis is $\pi/4$. When the plane wave is incident on the boundary of a perfect dielectric, the standing wave ratio in free space is $S=3$. If the wavelength of the transmitted wave in the perfect dielectric is a quarter of the wavelength in free space and the standing wave is minimal on the perfect dielectric boundary, solve the perfect dielectric's permittivity ϵ_1 and permeability μ_1 . [12]



Continued overleaf

Figure Q4

Appendix

Cylindrical coordinate system

$$\nabla \cdot \mathbf{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{e}_\rho & \rho \mathbf{e}_\phi & \mathbf{e}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

$$\nabla^2 u = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2}$$

Spherical coordinate system

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{e}_r & r \mathbf{e}_\theta & r \sin \theta \mathbf{e}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$$

$$\nabla^2 u = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2}$$

End of question paper